Paired t-Test: Line-by-Line Calculation

Before Training vs After Training

Person	Before	After	D = After – Before
1	62	70	8
2	68	74	6
3	71	78	7
4	75	80	5
5	64	71	7
6	80	85	5
7	73	79	6
8	69	75	6
9	77	83	6
10	66	72	6
11	72	78	6
12	81	87	6

1. Mean of Before Training

Mean = ΣX / n
Sum = 62 + 68 + 71 + 75 + 64 + 80 + 73 + 69 + 77 + 66 + 72 + 81 = 858
n = 12
Mean = 858 / 12 = **71.5**

2. Mean of After Training

Sum = 70 + 74 + 78 + 80 + 71 + 85 + 79 + 75 + 83 + 72 + 78 + 87 = 932
Mean = 932 / 12 = **77.6667**

3. Calculate Paired Differences

D = After – Before
D values = 8, 6, 7, 5, 7, 5, 6, 6, 6, 6, 6, 6

4. Sum of Differences

ΣD = 8 + 6 + 7 + 5 + 7 + 5 + 6 + 6 + 6 + 6 + 6 + 6 = **74**

5. Mean Difference

D■ = ΣD / n = 74 / 12 = **6.1667**
Average improvement = 6.17 points.

6. Difference Variance

s^2D = Σ(D – D■)^2 / (n – 1)
D■ = 6.1667
Σ(D – D■)^2 = 7.6667
n – 1 = 11
s^2D = 7.6667 / 11 = **0.69697**

7. Standard Deviation of Differences

sD = √0.69697 = **0.83485**

8. Standard Error

SE = sD / √n
SE = 0.83485 / √12
SE = 0.83485 / 3.4641 = **0.2410**

9. t-Statistic

t = (D■ – μD) / SE
Hypothesized Mean Difference μD = 0
t = (6.1667 – 0) / 0.2410 = **25.5879**
Excel shows –**25.5879** because it uses Before – After. The sign does not change the two-tailed conclusion.

10. Degrees of Freedom

df = n – 1 = 12 – 1 = **11**

11. Pearson Correlation

Pearson Correlation = **0.994132375**.
This is a very strong positive relationship between Before and After scores.

12. P-Values

One-tail p-value = **1.87457E–11**
Two-tail p-value = **3.74915E–11**

13. Critical t-Value

For α = 0.05 and df = 11:
Two-tail t Critical = **2.20098516**

14. Final Decision

|t Stat| = 25.5879
t Critical = 2.200985
25.5879 > 2.200985 → **Reject H■**
Also, 3.74915E–11 < 0.05 → **Reject H■**

15. Final Conclusion

There is a **statistically significant difference** between Before Training and After Training scores.

Before mean = 71.5
After mean = 77.6667
Average improvement = 6.1667 points.

Therefore, the training produced a statistically significant improvement in scores.

Quick Formula Summary

Step	Formula	Value
1	D = After – Before	Differences
2	$D\blacksquare = \Sigma D / n$	6.1667
3	$s^2D = \Sigma(D - D\blacksquare)^2 / (n - 1)$	0.69697
4	$sD = \sqrt{}s^2D$	0.83485
5	$SE = sD / \sqrt{}n$	0.2410
6	$t = (D\blacksquare - 0) / SE$	25.5879
7	$df = n - 1$	11